
Counting and conditional probability

Events · Counting · Bayes' theorem

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Probability and Statistics for Engineers

September 28, September 30, and October 2, 2026

Course support and deadlines

Ed Discussion: questions, announcements, and private messages. **No Canvas Inbox.**

[Open Ed from Canvas](#) All times Pacific

[Full contact details and meeting links](#)

Instructor / TAs	Office hours (Zoom)	Optional, ungraded discussions
Antonio de Leon	Wed 4–5 p.m.	—
Jeannine Romero	Tue / Thu 10–11 a.m.	01A: Fri 9:20–10:25 a.m. R Carson Acad 240 (in person)
Ehsan Mirafzali	Mon 3–5 p.m.	01B: Fri 10:40–11:45 a.m. (Zoom)
Seokjun Choi	Wed 3–4 p.m. 5:05–6:05 p.m.	01C: Wed 4–5:05 p.m. (Zoom)

Upcoming deadlines

HW and quizzes: **10 p.m. Pacific**

Quiz 1	Sun, Oct 4	Canvas
HW 1	Tue, Oct 6	Gradescope

Office-hour times link to Zoom. Check Ed for meeting changes.

HW: 24-hour grace period. Quizzes: no grace period. Approved accommodations apply.



Two coin tosses

Experiment: toss a coin twice and record the results in order.

Write H for heads and T for tails.

An **outcome** is one complete result, such as HT .

The **sample space** Ω contains all possible outcomes:



$$\Omega = \{HH, HT, TH, TT\}. \quad HT \neq TH$$

An **event** is a set of outcomes. It occurs when the observed outcome belongs to that set.

Which outcomes give exactly one head?

Write the event E as a set.

Events in words and symbols

First toss heads: $A = \{HH, HT\}$. Second toss heads: $B = \{HH, TH\}$.

Meaning	Notation	Outcomes
Both A and B (intersection)	$A \cap B$	$\{HH\}$
A or B , including both (union)	$A \cup B$	$\{HH, HT, TH\}$
Outside A (complement)	A^c	$\{TH, TT\}$
In B but outside A (difference)	$B \setminus A$	$\{TH\}$

The **empty event** $\emptyset$ contains no outcomes.

Events are **disjoint** when they share no outcomes: $A \cap B = \emptyset$.

Check: is “exactly one head” the same as $A \cup B$?

Kolmogorov's probability axioms

Andrey Kolmogorov, 1933.

A probability model assigns a number $P(A)$ to each event A .

These numbers must satisfy three axioms:

1. **Nonnegativity** $P(A) \geq 0$ for every event A .

2. **Normalization** $P(\Omega) = 1$.

3. **Additivity for disjoint events**

If no two events share an outcome, then

$$P(A_1 \cup A_2 \cup \dots) = P(A_1) + P(A_2) + \dots$$

The third axiom applies to finite or countably infinite collections.

The empty event and the complement

Use additivity on disjoint pieces of Ω .

1. The empty event

The disjoint union $\Omega \cup \emptyset$ equals Ω , so

$$1 = P(\Omega \cup \emptyset) = P(\Omega) + P(\emptyset) = 1 + P(\emptyset).$$

Subtracting 1 gives $P(\emptyset) = 0$.

2. The complement of A

A and A^c are disjoint, and $A \cup A^c = \Omega$. Therefore,

$$1 = P(\Omega) = P(A \cup A^c) = P(A) + P(A^c).$$

Rearranging gives $P(A^c) = 1 - P(A)$.

Nested events and probability bounds

$A \subseteq B$ means every outcome in A also belongs to B .

The part of B outside A is $B \setminus A$.

$B = A \cup (B \setminus A)$ is a disjoint union. By additivity,

$$P(B) = P(A \cup (B \setminus A)) = P(A) + P(B \setminus A).$$

Subtract the smaller event

$$P(B \setminus A) = P(B) - P(A) \geq 0.$$

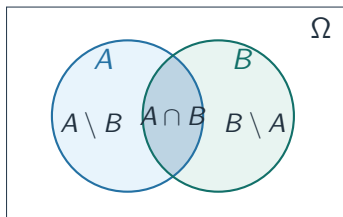
The last inequality is nonnegativity. Therefore,

$$A \subseteq B \implies P(A) \leq P(B) \quad (\text{monotonicity}).$$

Since $\emptyset \subseteq A \subseteq \Omega$, we have $0 = P(\emptyset) \leq P(A) \leq P(\Omega) = 1$.

Also, $A \cap B$ is inside both A and B , so $P(A \cap B) \leq \min\{P(A), P(B)\}$.

The union rule



1. Add only the part of B outside A .

$$P(A \cup B) = P(A) + P(B \setminus A).$$

2. Remove the overlap from B .

Since $B \setminus A = B \setminus (A \cap B)$, nested subtraction gives

$$P(B \setminus A) = P(B) - P(A \cap B).$$

Substitute: the overlap is counted once

$$P(A \cup B) = P(A) + P(B) - P(A \cap B).$$

If A and B are disjoint: $P(A \cup B) = P(A) + P(B)$.

Union bound: subtracting a nonnegative overlap gives

$$P(A \cup B) \leq P(A) + P(B).$$

Why counting helps with probability

Suppose Ω is **finite**, with $N = |\Omega|$ **equally likely** outcomes.

The notation $|A|$ means the number of outcomes in event A .

Add the probabilities of disjoint single-outcome events

For each outcome ω , the event $\{\omega\}$ has the same probability p .

$$1 = P(\Omega) = \sum_{\omega \in \Omega} P(\{\omega\}) = Np, \quad p = \frac{1}{N}.$$

For any event $A \subseteq \Omega$,

$$P(A) = \sum_{\omega \in A} P(\{\omega\}) = |A|p = \boxed{\frac{|A|}{|\Omega|}}.$$

The sum adds one term for each indicated outcome.

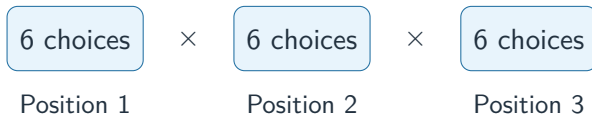
Our task: count A and Ω using the same kind of outcome.

A finite sample space alone does not guarantee equally likely outcomes.

Count one choice at a time

A robot sends a three-symbol code using six available symbols.

Order matters, and symbols may repeat.



Each first symbol has six continuations; each pair has six more.

Total: $6 \cdot 6 \cdot 6 = 6^3 = 216$ codes.

Multiplication principle for counting

If stage i has n_i choices after *every* allowed earlier sequence, then the number of complete sequences is $n_1 n_2 \cdots n_k$.

k positions with n choices each, repetition allowed: n^k sequences.

Ordered selections without replacement

From ten people, assign a president, a secretary, and a treasurer.
Each person can hold only one role. Switching roles changes the assignment.

President	Secretary	Treasurer
10 choices	9 remaining	8 remaining

Factorial notation: $n! = n(n-1) \cdots 1$, with $0! = 1$.

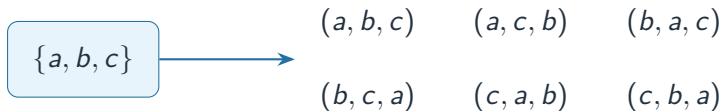
Choose k distinct objects from n , in order

$$(n)_k = n(n-1) \cdots (n-k+1) = \frac{n!}{(n-k)!}.$$

For integers $0 \leq k \leq n$. Ordering all n objects gives $n!$ permutations; choosing none gives $(n)_0 = 1$.

One committee, six orders

A committee has no roles. Reordering its members does not change the committee.



Why divide by 3!?

Each three-person committee appears in $3 \cdot 2 \cdot 1 = 6$ ordered lists.

The same reasoning gives $k!$ orders for each k -element subset.

Division works because every subset is counted the same number of times.

Unordered selections: combinations

Ten people, three committee members, no roles.

Divide the ordered count by the six orders of each committee.

Combinations: “ n choose k ”

$$\binom{n}{k} = \frac{\overbrace{\binom{n}{k}}^{\text{ordered selections}}}{\underbrace{k!}_{\text{orders of each subset}}} = \frac{n!}{k!(n-k)!}.$$

Here n, k are integers and $0 \leq k \leq n$.

Two checks: $\binom{n}{0} = 1$ (choose nothing) and $\binom{n}{n} = 1$ (choose everyone).

Three letters, three counting rules

Choose two letters from $\{A, B, C\}$. Write AB for A first, then B .

Selection rule	All possible outcomes	Count
Permutations Repetition allowed	$AA, AB, AC,$ $BA, BB, BC,$ CA, CB, CC	$3^2 = 9$
Permutations No repetition	$AB, AC, BA,$ BC, CA, CB	$(3)_2 = 3 \cdot 2 = 6$
Combinations No repetition	$\{A, B\}, \{A, C\}, \{B, C\}$	$\binom{3}{2} = \frac{6}{2!} = 3$

Permutations are ordered selections: $AB \neq BA$.

Combinations are subsets: $\{A, B\} = \{B, A\}$.

Four jobs, six servers

Assign four labeled jobs to six servers.

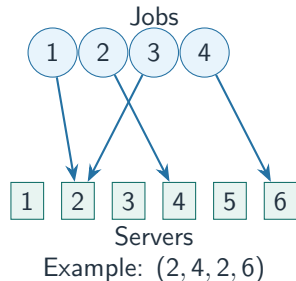
Model assumptions

- Every job can be assigned to any of the six servers.
- All complete assignments are equally likely.

Record servers in job order:

$$\omega = (x_1, x_2, x_3, x_4).$$

Each $x_i \in \{1, \dots, 6\}$. Repetition is allowed, so $|\Omega| = 6^4$.



What is the probability of a collision?

A collision means at least two jobs choose the same server.

A collision or no collision

Count the easier event: all four jobs use different servers.

$C = \{\text{assignments with at least two jobs on the same server}\},$

$C^c = \{\text{assignments with all four jobs on different servers}\}.$

All outcomes

Six choices per job.

No collision

6, 5, 4, then 3 choices.

Use equal probabilities, then take the complement.

Practice: at least one marked item

Choose a three-item subset from ten distinct items, with all such subsets equally likely. Four items are marked; six are unmarked.



What is the probability of choosing at least one marked item?

One outcome is an unordered three-item subset. Try counting “none marked.”

Choose the right counting method

First describe **one outcome**. Then ask: does order matter, and can objects repeat?

Outcome	Repetition	Count
Ordered length- k sequence	Allowed	n^k
Ordered length- k sequence	Not allowed	$(n)_k = n!/(n - k)!$
Unordered k -element subset	Not allowed	$\binom{n}{k} = n!/[k!(n - k)!]$

$n \geq 1$ and $k \geq 0$ are integers. Without replacement, $k \leq n$.

Turn the count into a probability

For equally likely outcomes, $P(A) = |A|/|\Omega|$.

Count both numerator and denominator with the same kind of outcome.

If “at least one” is hard to count: try “none,” then use the complement.

Wednesday: probability recap

Result	Statement
Empty event and sample space	$P(\emptyset) = 0, \quad P(\Omega) = 1$
Complement	$P(A^c) = 1 - P(A)$
Nested events	$A \subseteq B \implies P(A) \leq P(B)$
Intersection	$P(A \cap B) \leq \min\{P(A), P(B)\}$
Remove part of B	$P(B \setminus A) = P(B) - P(A \cap B)$
Union	$P(A \cup B) = P(A) + P(B) - P(A \cap B)$

Checks: $0 \leq P(A) \leq 1$ and $P(A \cup B) \leq P(A) + P(B)$.

For nested events, $A \cap B = A$, so subtraction becomes $P(B) - P(A)$.

Wednesday: counting recap

Selection of k objects from n	Count
Ordered, repetition allowed	n^k
Ordered, no repetition	$(n)_k = n!/(n - k)!$
Unordered, no repetition	$\binom{n}{k} = (n)_k/k!$

With equally likely outcomes, $P(A) = |A|/|\Omega|$.

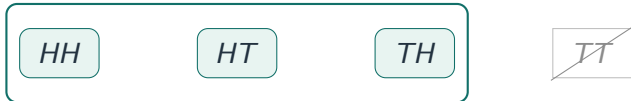
Recall: in two tosses with four equally likely ordered outcomes, what is the probability of exactly one head?

Today: how does the answer change when we learn that an event occurred?

Conditional probability

Use the coin model with four equally likely ordered outcomes.

Given at least one head, how likely is exactly one head?



Condition: $B = \{HH, HT, TH\}$. Event of interest: $A = \{HT, TH\}$.

Keep the outcomes in B ; rescale their probabilities to total 1

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}, \quad P(B) > 0.$$

With equally likely outcomes, $P(A \mid B) = \frac{|A \cap B|/|\Omega|}{|B|/|\Omega|} = \frac{|A \cap B|}{|B|}$.

Conditioning with unequal probabilities

A request is sent to one of three servers with the probabilities below.

Given that it did not go to server 1, how likely is server 2?

Server	Original probability	Given server 2 or 3
1	1/2	0 (excluded)
2	1/3	_____
3	1/6	_____

$\Omega = \{1, 2, 3\}$; keep $B = \{2, 3\}$. The question asks for $A = \{2\}$.

Count retained outcomes only when their probabilities are equal.

Two dice: at least one is four

		Second die					
		1	2	3	4	5	6
First die	1	1, 1	1, 2	1, 3	1, 4	1, 5	1, 6
	2	2, 1	2, 2	2, 3	2, 4	2, 5	2, 6
	3	3, 1	3, 2	3, 3	3, 4	3, 5	3, 6
	4	4, 1	4, 2	4, 3	4, 4	4, 5	4, 6
	5	5, 1	5, 2	5, 3	5, 4	5, 5	5, 6
	6	6, 1	6, 2	6, 3	6, 4	6, 5	6, 6

Model: all 36 ordered dice outcomes are equally likely.

Given at least one die is 4, how likely is a sum of 7?

$A = \{\text{sum is 7}\}.$

$B = \{\text{at least one die is 4}\}.$

Outlined row and column: keep B .

Circled cells: $A \cap B$.

Different conditions, different probabilities

Keep the same equally likely dice outcomes and $A = \{\text{sum is } 7\}$.

Condition B	$ B $	$ A \cap B $	$P(A B)$
At least one die is 4	11	2	2/11
First die is 4	_____	_____	_____
Exactly one die is 4	_____	_____	_____

Explain the differences

- Which condition excludes $(4, 4)$?
- Which condition excludes $(3, 4)$?

The conditional product rule

Multiply the defining ratio by $P(B) > 0$:

$$P(B) P(A | B) = P(B) \frac{P(A \cap B)}{P(B)} = P(A \cap B).$$

Probability of two events occurring together

$$P(A \cap B) = P(B) P(A | B).$$

Dice example: A = sum of 7; B = at least one die is 4.

$$\underbrace{\frac{11}{36}}_{P(B)} \times \underbrace{\frac{2}{11}}_{P(A|B)} = \underbrace{\frac{2}{36}}_{P(A \cap B)}.$$

If $P(A) > 0$, reversing the roles also gives $P(A \cap B) = P(A) P(B | A)$.

Complements within a condition

Assume $P(B) > 0$. Within B , either A occurs or A^c occurs.

$$B = (A \cap B) \cup (A^c \cap B) \quad (\text{disjoint pieces}).$$

Add their probabilities, then divide by $P(B)$:

$$1 = \frac{P(B)}{P(B)} = \frac{P(A \cap B) + P(A^c \cap B)}{P(B)} = P(A \mid B) + P(A^c \mid B).$$

Conditional complement rule

$$P(A^c \mid B) = 1 - P(A \mid B).$$

Coin example: given at least one head, exactly one head has probability $2/3$.

The remaining case is HH , with probability $1 - 2/3 = 1/3$.

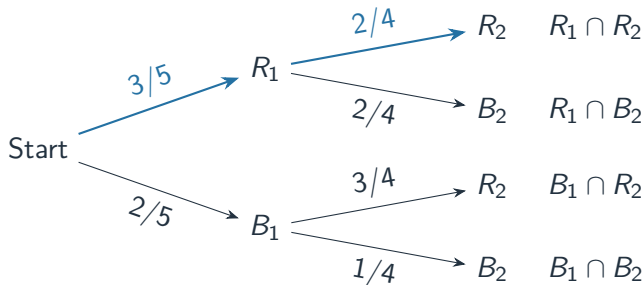
The condition B stays the same throughout. Also, $P(B \mid B) = P(B)/P(B) = 1$.

Two draws without replacement

A bag contains three red and two blue balls. Draw twice without replacement.

At each draw, every remaining ball is equally likely.

Let R_i mean red on draw i and B_i mean blue on draw i .



Second-draw branches use the remaining balls. The product rule gives a complete path's probability.

Two red draws

What is the probability that both draws are red?

	Before drawing	After one red draw
Red balls	3	2
All remaining balls	5	4
Chance of red next	$3/5$	$2/4$

The second probability describes the remaining balls

After one red draw, two red balls remain among four balls: $P(R_2 \mid R_1) = 2/4$.

Three draws and the chain rule

Assume $P(A \cap B) > 0$.

Apply the product rule twice

$$\begin{aligned} P(A \cap B \cap C) &= P(A \cap B) P(C \mid A \cap B) \\ &= P(A) P(B \mid A) P(C \mid A \cap B). \end{aligned}$$

What is the probability of red, then red, then blue?

Use the same bag and draw three times without replacement.

First: red	After red: red	After two reds: blue
3 red of 5 balls	2 red of 4 balls	2 blue of 3 balls

Each factor uses all the earlier information.

Practice: both draws or a given first draw

Use the same bag: three red and two blue balls; no replacement.

Two different questions

1. What is the probability that the two draws have different colors?
2. Given the first draw is blue, what is the probability the second is red?

“Given blue first” starts after that draw. “Blue then red” includes reaching it.

Conditional probability summary

For $P(B) > 0$

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}, \quad P(A \cap B) = P(B) P(A \mid B).$$

1. Name the known event B in words.
2. Find the outcomes in both A and B .
3. Divide by the probability of the condition B .

Direction matters

$P(A \mid B)$ and $P(B \mid A)$ usually answer different questions.

Equal outcome probabilities allow counts. Otherwise, use their probability weights.
Friday: combine disjoint cases, then reverse the conditioning.

Friday: conditional probability recap

For a condition B with $P(B) > 0$:

Question	Rule
A given B	$P(A \mid B) = P(A \cap B) / P(B)$
A and B together	$P(A \cap B) = P(B) P(A \mid B)$
Not A , given B	$P(A^c \mid B) = 1 - P(A \mid B)$

For successive draws, multiply along a path using all earlier results as conditions.

Recall: three red and two blue balls, two draws without replacement.

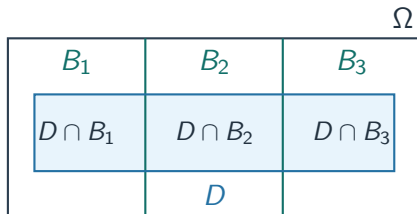
What is the probability of two red draws?

Today: combine disjoint cases, then reverse the direction of conditioning.

Partitions

Each device comes from exactly one of three suppliers.

Let $B_i = \{\text{supplier } i\}$ and $D = \{\text{device is defective}\}$.



Schematic: region areas do not represent probabilities.

No overlap

A device has only one supplier.

No missing cases

Every device has a supplier.

These cases form a **partition** of Ω .

Definition

$B_1, \dots, B_m$ partition Ω when $B_i \cap B_j = \emptyset$ for $i \neq j$
and $B_1 \cup \dots \cup B_m = \Omega$.

The law of total probability

A defect can come from supplier 1, 2, or 3. How do we combine these cases?

$D = (D \cap B_1) \cup (D \cap B_2) \cup (D \cap B_3)$ is a disjoint union.

$$\begin{aligned} P(D) &= P(D \cap B_1) + P(D \cap B_2) + P(D \cap B_3) \\ &= P(D \mid B_1) P(B_1) + P(D \mid B_2) P(B_2) \\ &\quad + P(D \mid B_3) P(B_3). \end{aligned}$$

General rule: sum over a partition

$$P(A) = \sum_{i=1}^m P(A \cap B_i) = \sum_{i=1}^m P(A \mid B_i) P(B_i).$$

Assume the B_i partition Ω and $P(B_i) > 0$.

The sum adds one weighted contribution for each case.

Parts with probability zero contribute zero and can be left out of the conditional sum.

Defects from three suppliers

Select a device at random. Its supplier shares and defect rates are:

Supplier	Share $P(B_i)$	Defect rate $P(D B_i)$	$P(D \cap B_i)$
1	0.20	0.01	_____
2	0.30	0.02	_____
3	0.50	0.03	_____

What is the probability that the selected device is defective?

Use the supplier shares as weights: they do not supply equal numbers of devices.

Bayes' theorem

Given an observed event A , how likely is case B_j ?

Start with the conditional definition and expand its numerator:

$$P(B_j | A) = \frac{P(A \cap B_j)}{P(A)} = \frac{P(A | B_j) P(B_j)}{P(A)}.$$

Use total probability for the denominator

$$P(B_j | A) = \frac{P(A | B_j) P(B_j)}{\sum_{i=1}^m P(A | B_i) P(B_i)}.$$

Numerator

Case B_j and observation A occur together.

B_i form a partition; $P(B_i) > 0$ and $P(A) > 0$.

$P(B_j)$ is the prior; $P(B_j | A)$ is the posterior, after observing A .

Denominator

All ways the observation A can occur.

Which supplier made the defective device?

Given a defect, what is the probability the device came from supplier 3?

The previous contributions, shown on a common scale of 1,000 devices:

	Supplier 1	Supplier 2	Supplier 3
All devices	200	300	500
Defective devices	2	6	15

Counts represent the model proportions, not a guaranteed observed sample.

Restrict to defective devices. There are $2 + 6 + 15 = 23$ on this scale.

Before observing the defect: $P(B_3) = 50\%$.

After observing it: compare supplier 3 with all defective devices.

A rare issue and an alert

A monitoring system checks devices for a fault. Suppose the probabilities are:

$G = \{\text{genuine fault}\}$; $L = \{\text{alert}\}$.

What the model tells us	Probability
A device has a fault	$P(G) = 0.01$
Alert, given a fault	$P(L \mid G) = 0.90$
Alert, given no fault	$P(L \mid G^c) = 0.05$

Given an alert, how likely is a genuine fault?

We want $P(G \mid L)$. This reverses the condition in $P(L \mid G)$.

Count the devices with an alert

Show the model proportions on a scale of 10,000 devices.

Alerts: $100(0.90) = 90$ among faulty devices; $9,900(0.05) = 495$ among the rest.

	Alert L	No alert L^c	Total
Fault G	90	10	100
No fault G^c	495	9,405	9,900
Total	585	9,415	10,000

Given an alert: keep the alert column. Which of those devices have a fault?

The fault rate is only 1%. The much larger no-fault group produces many false alerts.

Practice: which box was chosen?

First choose a box. Then choose a ball, with its four balls equally likely.

	Chance of choosing box	Contents	Chance of red
Box 1	1/3	3 red, 1 blue	3/4
Box 2	2/3	1 red, 3 blue	1/4

Given a red draw, what is the probability we chose box 1?

Let $B_i = \{\text{box } i\}$ and $R = \{\text{red draw}\}$. We want $P(B_1 | R)$.

Total probability and Bayes: summary

Total probability: combine all cases

$$P(A) = \sum_{i=1}^m P(A \mid B_i) P(B_i).$$

Bayes: update a case after an observation

$$P(B_j \mid A) = \frac{P(A \mid B_j) P(B_j)}{P(A)}.$$

Before calculating

- The cases must be disjoint and cover the sample space.
- Use each case's probability as its weight.
- Conditioning events must have positive probability.

Next week: independence—when does conditioning leave a probability unchanged?

References

DeGroot and Schervish

Probability and Statistics, fourth edition.

- Sections 1.4–1.8 and 1.10: sets, probability rules, counting, and unions.
- Section 2.1: conditional probability, product rule, and total probability.
- Section 2.3: Bayes' theorem; Example 2.3.4 motivates the supplier example.

Mor Harchol-Balter

Introduction to Probability for Computing (2024).

- Section 1.5: permutations and combinations.
- Sections 2.1–2.3, 2.5–2.6: events, probability, conditioning, and Bayes.

Additional background: Sheng Jiang, *STAT 131: Introduction to Probability Theory*, July 18, 2024, pp. 10–19; Antonio de Leon's earlier STAT 131 discussion sheets and recap slides. Textbook reading is complementary; slides and course notes are the primary materials.